

LAPORAN TUGAS 3: IMPLEMENTASI DATA WAREHOUSE PEMROSESAN DAN INFRASTRUKTUR DATA KELAS C



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Bukti Instalansi Dependency

Bukti verifikasi dependency environment

```
Dependency environment verification
Interpreter: /workspaces/.venv/bin/python

pandas: available (version 3.0.6)
numpy: available (version 2.4.6)
sqlalchemy: available (version 2.1.1)
matplotlib: available (version 3.11.2)
jupyter: available (version installed)
```

Bukti Eksekusi Notebook ETL

Screenshot proses ETL (eksekusi notebook)

Diambil otomatis dari environment dan artefak proyek.

```
Notebook: hands_on_data_warehouse.ipynb
Total code cells: 27
Executed code cells: 27
Unexecuted code cells: 0
Status: SUCCESS
```

Bukti Validasi Tabl Warehouse

Screenshot validasi tabel warehouse

Diambil otomatis dari environment dan artefak proyek.

```
dim_sensors: 10
dim_locations: 5
dim_time: 1440
fact_sensor_readings: 14400
```

```
Schema tables:
- dim_locations
- dim_sensors
- dim_time
- fact_sensor_readings
- sqlite_sequence
```

Hasil Query Analitik

Screenshot hasil 3 query analitik

Diambil otomatis dari environment dan artefak proyek.

[Query 1] Rata-rata AQI per lokasi (terbaik 5)

```
('Location_D', 57.36, 2880)
('Location_C', 57.67, 2880)
('Location_B', 57.72, 2880)
('Location_E', 57.78, 2880)
('Location_A', 58.1, 2880)
```

[Query 2] Rata-rata AQI per periode waktu

```
('Night', 50.6, 4200)
('Evening', 56.27, 3000)
('Afternoon', 60.05, 3600)
('Morning', 64.92, 3600)
```

[Query 3] Jumlah pembacaan per sensor (top 10)

```
('SENSOR_001', 1440)
('SENSOR_002', 1440)
('SENSOR_003', 1440)
('SENSOR_004', 1440)
('SENSOR_005', 1440)
('SENSOR_006', 1440)
('SENSOR_007', 1440)
('SENSOR_008', 1440)
('SENSOR_009', 1440)
('SENSOR_010', 1440)
```

TERMINAL VERIFICATION – DATA WAREHOUSE

Evidence generated from the verified SQLite warehouse

```
$ pwd

/workspaces/PID_etl/Load-data ke warehouse

$ python3 -c "verifikasi database warehouse"

DATABASE: /workspaces/PID_etl/Load-data ke warehouse/warehouse/sensor_warehouse.db
DATABASE_EXISTS: True
DATABASE_SIZE_MB: 3.73
--- TABLE COUNTS ---
dim_sensors: 10
dim_locations: 5
dim_time: 1440
fact_sensor_readings: 14400
--- ANALYTIC RESULTS ---
best_location: Location_D
highest_aqi_location: Location_A
best_time_period: Night
sensor_readings:
  SENSOR_001: 1440
  SENSOR_002: 1440
  SENSOR_003: 1440
  SENSOR_004: 1440
  SENSOR_005: 1440
  SENSOR_006: 1440
  SENSOR_007: 1440
  SENSOR_008: 1440
  SENSOR_009: 1440
  SENSOR_010: 1440
--- SOURCE DATA ---
csv_rows: 28814
csv_columns: 25
duplicates_after_location_standardization: 14414
unique_rows_after_cleaning: 14400
--- STATUS ---
VERIFICATION: PASSED
```

Beberapa Grafik Hasil Kode





